Feraz Rab

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Actively seeking full-time opportunities • May 2027 • Fabs, Foundries, OEMs, OSAT/ATMP & Semiconductor R&D

PROFESSIONAL SUMMARY

Materials engineer with a dual foundation in **Semiconductor Technology** (M.Tech, IISc–NTU Joint Degree) and **Metallurgical & Materials Engineering** (B.Tech, Gold Medal — NIT Durgapur). Combines hands-on plasma deposition and thin-film process experience with computational and data-driven modeling skills — including **physics-based simulations (SEMulator3D, TCAD, Dislocation Dynamics)**, **Design of Experiments (DOE)**, and **statistical / ML methods** (Response Surface Modeling, Bayesian optimization, Gaussian Process regression) applied to process–property–performance problems. Experienced in bridging experimental characterisation data (SEM, XRD, UV-Vis, PL, TCSPC) with model outputs to close simulation–experiment gaps across thin-film deposition, device integration, and process optimisation workflows. Interested in roles spanning process/integration engineering, process modeling & simulation, and data-driven process development across the semiconductor manufacturing ecosystem.

EDUCATION

Indian Institute of Science (IISc), Bangalore *&* **National Taiwan University (NTU), Taiwan** — **July 2025 – May 2027**
M.Tech in Semiconductor Technology *(IISc–NTU Joint Degree)* — CGPA: 8.10 / 10

National Institute of Technology (NIT) Durgapur, West Bengal, India — **December 2020 – May 2024**
B.Tech in Metallurgical & Materials Engineering — CGPA: **9.35 / 10** ⋆ **Institute Gold Medal**

TECHNICAL SKILLS

Process Simulation & Computational Modeling: SEMulator3D (feature-scale virtual fabrication), Silvaco TCAD (device & process simulation), Dislocation Dynamics (DD Lab — MATLAB/Octave), Physics-Based Process Modeling, Ion/Radical Transport & Plasma–Surface Interaction Mechanisms, Feature-Scale & Reactor-Scale Simulation Concepts

AI / ML & Data-Driven Methods: Response Surface Modeling (RSM), Gaussian Process (GP) Regression, Bayesian Optimization, Uncertainty Quantification, Design of Experiments (DOE), Statistical Process Control (SPC), Probabilistic Machine Learning Concepts, Scientific Data Analysis & Visualisation (Python, MATLAB, OriginPro, JMP)

Plasma Deposition & Thin-Film Process Knowledge: RF/DC Magnetron Sputtering (PVD), Thermal & E-Beam Evaporation, PECVD / CVD, PEALD (concepts), Spin Coating & Annealing, Plasma Cleaning, Dry Etching (RIE/ICP), CMP, Process Parameter – Film Property Correlation, Deposition Recipe Development & Optimisation

Experimental Characterisation & Metrology: SEM/EDX, XRD (crystal structure & phase ID), XPS, TEM, AFM, UV-Vis Spectroscopy, Photothermal Deflection Spectroscopy (PDS), Steady-State PL, TCSPC, PLQY, FTIR, Four-Probe Resistivity, Simulation–Experiment Data Comparison & Gap Analysis

Software & Programming: Python (data analysis, scientific computing), MATLAB / Octave (simulations, scripting), SEMulator3D, Silvaco TCAD, OriginPro, JMP, MS Office Suite

EXPERIENCE

Graduate Engineer Trainee — *Hindalco Industries (Aditya Birla Group), Vadodara, India* — *August 2024 – June 2025*

- Applied **SPC, DOE, and Lean methods** to monitor and model process KPIs across the Copper Tube Greenfield Project; used structured data analysis to identify deviation root causes and implement corrective actions in a high-volume manufacturing environment.
- Led shift commissioning: end-to-end process validation, precision equipment calibration, and trial runs, building disciplined process documentation and cross-functional communication skills.
- Coordinated with engineering, quality, and production teams in a matrix environment, developing cross-functional collaboration and reporting skills.

Capacity Building Program in Semiconductor Manufacturing — *NTHU, Hsinchu, Taiwan & IIT Hyderabad, India* *June 2024 – July 2024*

- Gained hands-on process knowledge in PVD thin-film deposition, RF magnetron sputtering, PECVD, UV photolithography, dry/wet etching, and ion implantation in a live fab environment.
- Systematically correlated deposition parameters (power, pressure, gas flow) with measured film structural and electrical properties, practising parameter–output mapping central to process characterisation and modeling.
- Developed intuition for plasma–surface interactions and how process variables drive material outcomes across the front-end process flow.

Summer Research Intern — *Indian Institute of Technology Hyderabad, India* *May 2023 – July 2023*

- Investigated synthesis and multi-step functional modification of bacterial cellulose (BC): culture growth, mechanical and chemical modification of BC membranes, and synthesis of magnetic nanoparticles via co-precipitation.
- Fabricated and characterised BC-based magnetic composites and magnetic fluids; evaluated structural integrity and magnetic response via SEM and physical property measurements, practising multi-variable experimental design and quantitative data interpretation.

Summer Research Intern — *Indian Institute of Technology Bombay, India* *May 2022 – July 2022*

- Built a rigorous foundation in dislocation theory through graduate-level textbooks and a GIAN course on crystal plasticity, then implemented **Dislocation Dynamics (DD) simulations** using the DD Lab program (MATLAB / Octave) to model **Frank–Read source activation** and **dislocation junction formation**.
- Analysed simulation outputs to extract physically interpretable trends — linking atomic-scale interaction parameters to emergent macroscopic behaviour — developing a **physics-based computational modeling workflow** (set up → run → analyse → compare with theory).
- Prepared technical summaries of simulation results, practising the clear communication of complex modeling outputs to an academic audience.

KEY PROJECTS

Data-Driven Process–Property Modeling: 2D $(PA)_2PbI_4$ Perovskite Thin Films *Jan – Apr 2026*

- Fabricated $(PA)_2PbI_4$ thin films across a systematically varied concentration range (0.05–0.5 M) via spin coating and controlled annealing; treated precursor concentration as the primary **process input variable** to build a quantitative process–property model.
- Executed a multi-technique spectroscopic characterisation campaign — UV-Vis (bandgap), PDS (sub-bandgap trap states, Urbach energy), steady-state PL, **TCSPC** (carrier lifetime, radiative/non-radiative recombination rates), and **PLQY** (emission efficiency) — generating a rich experimental dataset for model validation.
- Correlated process inputs (concentration) with quantitative material outputs (trap density, recombination dynamics, emission efficiency), producing a **simulation-ready process–property map**.

Physics-Informed DOE Modeling: SnO_2 TCO Thin-Film Process Optimisation *Aug – Nov 2025*

- Designed and executed a **full-factorial DOE** for RF magnetron sputtered SnO_2 thin films, varying RF power and O_2 partial pressure; generated a structured experimental dataset mapping inputs to film microstructure, crystal orientation (XRD), and optical transmittance ($> 85\,\%$).
- Applied **Response Surface Modeling (RSM)** concepts to fit process parameter − film property relationships from small-sample experimental data, identifying the mechanistic drivers of performance.
- Compared measured film properties across DOE runs to identify process–model alignment and gaps, proposing physically grounded parameter refinements.

Simulation & Failure Analysis: W2W Bonding for 3D-IC Integration *Aug – Nov 2025*

- Modelled hybrid wafer-to-wafer bonding for 3D-IC integration: alignment tolerances, interfacial void nucleation, thermal stress distributions, and yield — building **feature-scale process modeling** intuition for multi-layer semiconductor stacks.
- Benchmarked simulation predictions against reported experimental results across flip-chip, die-to-wafer, and W2W bonding techniques, identifying parameter mismatches and proposing model refinements.

Additive Manufacturing & High-Entropy Alloys — Elsevier Book Chapter *2023 – 2024*

- Co-authored a peer-reviewed book chapter on AM of HEAs for ballistic impact applications (Elsevier, 2026), requiring synthesis of multi-physics process–structure –property relationships and clear technical communication of complex modeling concepts to a broad engineering audience.

PUBLICATIONS

Rab, F.; Saha, M. "Potential and Perspectives on Additive Manufacturing of High-Entropy Alloys for Ballistic Impact Applications." In *High-Entropy Alloys: Additive Manufacturing, Ballistic Impact Response and Dynamic Mechanical Applications*, Elsevier, Chapter 9, 2026. [View Publication]

AWARDS & HONOURS

- **IIM Student Prize (2024)** — Indian Institute of Metals; ranked among top three CGPAs across all NITs and universities in India in Metallurgical & Materials Engineering.
- **Institute Gold Medal** — Highest CGPA, Metallurgical & Materials Engineering, NIT Durgapur.
- **Smt. Tarulata Sinha Memorial Gold Medal** — Overall excellence and top departmental performance, NIT Durgapur.
- **Six Sigma Yellow Belt** — Kennesaw State University; structured experimental design, DOE, and statistical process optimisation.
- **OPJEMS Scholar (2022)** — Selected among top 80 engineering students in India by JSW Group for academic and leadership excellence.